

CSS Transition

- CSS transition is a module of CSS.
- Transition is used for configuring animation for initial and final state.
- Initial state is before transformation.
- Final State is after transformation.
- You can control the behaviour using various CSS effects
 - timing
 - duration
 - other attributes
- CSS Transition comprises of following properties
 - transition
 - transition-delay
 - transition-duration
 - transition-property
 - transition-timing-function

transition-delay	Delay is time taken to start animation. Syntax: transition-delay: 2s;
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transition-duration	Time take to complete the transformation. Specifies the duration from initial to final and vice versa. Syntax: transition-duration: 5s;
transition-property	-You can define several effects for transformation. -All effects are transformed from initial to final. -You can define only the effect that you want to transform by using transform-property. Syntax: transition-property: width, height; transition-property:all;
transition-timing-function	-CSS provides a set of pre-defined animations functions which you can apply to content. -These effects have pre-defined timings and behaviour.

	-The transition timing functions are - linear: speed will be even from begin to end. - ease-in: starts slowly and increases towards end. - ease-out: starts quickly and slow down. - ease-in-out - steps - jump-start - jump-end - jump-none - jump-both - start - end - step-start - step-end
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Ex: **Duration and Delay**

```
<!DOCTYPE html>
```

```
<html>
```

```
  <head>
```

```
    <title>Timing</title>
```

```
    <style>
```

```
img {
    margin-left: 10px;
    transition-duration: 2s;
    transition-delay: 5s;
}
img:hover {
    transform: translate(400px,0);
    transition-duration: 2s;
}
</style>
</head>
<body>
    
    </body>
</html>
```

Ex: **Transition Property**

```
<!DOCTYPE html>
<html>
    <head>
        <title>Timing</title>
        <style>
            div{
```

```
        width: 100px;
        height: 100px;
        background-color: red;
        transition-duration: 2s;
        transition-property: width, height;
    }
    div:hover {
        width: 300px;
        height: 300px;
        background-color: yellow;
    }
</style>
</head>
<body>
    <div>

        </div>
</body>
</html>
```

Ex: CSS Timing functions

```
<!DOCTYPE html>
<html>
    <head>
```

```
<title>Timing</title>
<style>
  div{
    width: 100px;
    height: 100px;
    transition-duration: 2s;
    background-color: red;
  }
  div:hover {
    width: 1000px;
  }
  #easein {
    transition-timing-function: step-start;
    transition-timing-function: steps(10);
  }
</style>
</head>
<body>
  <div id="easein">
```

Hello ! Welcome to CSS Effects and Animations

```
</div>  
</body>  
</html>
```

CSS Animations

- CSS animations are used to animate CSS transitions.
- You can reduce the use of JavaScript and Flash for animation.
- Animation can easily run on low bandwidth devices like mobiles, tabs.
- Animation will work even when JavaScript is disabled or flash plugin is not available.
- To create CSS animation, we need
 - @keyframes
 - animation
 - animation-name
 - animation-duration
 - animation-timing-function
 - animation-delay

- animation-iteration-count
- animation-direction
- animation-fill-mode
- animation-play-state

@Keyframes

- It is used to configure every frame in animation.
- At each frame we can define effects for object.
- Keyframes are defines with initial, final, and intermediate steps.

Syntax:

```
@keyframes identifierName
{
    from { effects for initial state }
    to { effects for final state }
}
```

Syntax:

```
@keyframes identifier
{
    0% { effects }
    40% { effects }
```



```
80% { effects }  
100% {effects}  
}
```

- Keyframes are applied to any element by using “animation-name” property

Syntax:

```
P {  
    animation-name:  
    keyframeIdentifierName;  
}
```

EX:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Timing</title>
```

```
<style>
```

```
.container {
```

```
    overflow: hidden;
```

```
}
```

```
@keyframes slideIn
```

```
{
```

```
    from
    {
        margin-left: 100%;
        transform: scale(3);
    }
    to
    {
        margin-left: 0%;
        transform: scale(1);
    }
}
h2{
    animation-name: slideIn;
    animation-duration: 5s;
}
</style>
</head>
<body class="container">
    <h2>CSS Animations</h2>
```

```
</body>
```

```
</html>
```

- You can add frames in between the start and end by using step percentage.

Ex:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Timing</title>
```

```
<style>
```

```
.container {
```

```
    overflow: hidden;
```

```
}
```

```
@keyframes slideIn
```

```
{
```

```
    from
```

```
{
```

```
    margin-left: 100%;
```

```
    transform: scale(1);
```

```
    }
    50% {
        font-size: 300%;
    }
    to
    {
        margin-left: 0%;
        transform: scale(1);
    }
}
h2{
    animation-name: slideIn;
    animation-duration: 5s;
}
</style>
</head>
<body class="container">
    <h2>CSS Animations</h2>
</body>
```

</html>

- You can control the iteration count, the number of times to display by using **“animation-iteration-count”**

Ex:

<!DOCTYPE html>

<html>

<head>

<title>Timing</title>

<style>

.container {

overflow: hidden;

}

@keyframes slideIn

{

from

{

margin-left: 100%;

transform: scale(1);

```
    }
    50% {
        font-size: 300%;
    }
    to
    {
        margin-left: 0%;
        transform: scale(1);
    }
}
h2{
    animation-name: slideIn;
    animation-duration: 5s;
    animation-iteration-count: infinite;
}
</style>
</head>
<body class="container">
    <h2>CSS Animations</h2>
```

</body>

</html>

- You can control animation direction by using “animation-direction”

Ex

```
<!DOCTYPE html>
```

```
<html>
```

```
  <head>
```

```
    <title>Timing</title>
```

```
    <style>
```

```
      .container {
```

```
        overflow: hidden;
```

```
      }
```

```
      @keyframes slideIn
```

```
      {
```

```
        from
```

```
        {
```

```
          margin-left: 100%;
```

```
          transform: rotate(360deg);
```

```
        }
```

```
        to
```

```
        {
```

```
        margin-left: 0%;
        transform: rotate(0deg);
    }
}

img {
    width: 20%;
    height: 70%;
    animation-name: slideIn;
    animation-duration: 5s;
    animation-iteration-count: infinite;
    animation-direction: alternate;
    animation-delay: 5s;
    animation-timing-function: ease;
}
</style>
</head>
<body class="container">
    
    </body>
</html>
```


